

Koiter asymptotic analysis about a non-linear pre-buckling state

Implementation notes for `NLprebuck=True` in `bfscylinder_models`

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Abstract

This note documents the implementation of the single-mode Koiter method with a non-linear pre-buckling state in `koiter_cylinder_newton_raphson.py` (Donnell-Mushtari-Vlasov kinematics) and `koiter_cylinder_newton_raphson_sanders.py` (Sanders-Koiter kinematics), and of the same algorithm in the variable-stiffness models `koiter_cylinder_CTS.py` and `koiter_cylinder_CTS_sanders.py` (Section 12). It covers the tangent stiffness matrix and the consistency it must satisfy, the reduction of the pre-buckling problem to the axisymmetric subspace, the incremental-iterative scheme that places the expansion point in the neighbourhood of the bifurcation point, the removal of the arbitrariness left by the degenerate buckling modes, and the stiffness-weighted orthogonality condition imposed on the second-order fields, and the mesh resolution the continuous-tow-shearing cylinders require. The notation follows the finite element chapter of Jansen et al. [4]. Equation numbers of the form “Sun Eq. (nm)” refer to Sun et al. [6].

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1 Scope and notation

The routine `fkoiter_cyl_SS3` performs, for a circular cylindrical shell under axial compression with SS-3 boundary conditions,

1. a pre-buckling analysis,
2. a buckling eigenvalue analysis about the pre-buckling state,
3. the Koiter asymptotic expansion about the bifurcation point, returning the post-buckling coefficients a_{ijk} and b_{ijkl} .

With `NLprebuck=False` the pre-buckling state is the linear elastic solution and the pre-buckling path is exactly linear in the load parameter. With `NLprebuck=True` the pre-buckling state is obtained from a geometrically non-linear analysis, and the whole expansion must then be carried out about a state *in the neighbourhood of the bifurcation point*. This document describes the latter case.

The shell coordinates are the axial x and the circumferential y , the radius is r and the thickness h . Following dos Santos and Castro [2], y is the circumferential coordinate measured as an *arc length*, $y = r\vartheta$ with ϑ the polar angle, so that x and y have the same units and the membrane strains below carry no explicit factor of r . The symbol θ is reserved throughout for the tow steering angle of Section 12. The circumferential discretisation is n_y elements around the circumference, the axial one n_x stations along the generator. The degrees-of-freedom are collected in $\bar{\mathbf{u}}$, the fundamental (pre-buckling) state is $\hat{\mathbf{u}}$, and the buckling modes are $\delta\bar{\mathbf{u}}_i$. The external load is proportional, $\mathbf{F}_{ext} = \lambda\mathbf{F}_{ref}$, with $\mathbf{F}_{ref}$ the reference load vector built from N_{xx}^{unit} . Two load levels are distinguished: λ_b , at which the pre-buckling state is evaluated, and λ_c , the bifurcation load.

2 Kinematics

The membrane strains are written, for both kinematic models, as

$$\boldsymbol{\varepsilon} = \mathbf{B}_m \bar{\mathbf{u}} + \frac{1}{2} \begin{Bmatrix} g_x^2 \\ g_y^2 \\ 2g_x g_y \end{Bmatrix}, \quad \boldsymbol{\kappa} = \mathbf{B}_b \bar{\mathbf{u}} \quad (1)$$

where g_x and g_y are the two gradient measures that carry the geometric non-linearity:

$$\text{Donnell-Mushtari-Vlasov: } g_x = w_{,x}, \quad g_y = w_{,y}, \quad \text{Sanders-Koiter: } g_x = w_{,x}, \quad g_y = w_{,y} - \frac{v}{r} \quad (2)$$

Every non-linear term of the Sanders-Koiter membrane strains is recovered from the Donnell-Mushtari-Vlasov one by the substitution $(w_{,x}, w_{,y}) \rightarrow (g_x, g_y)$, which is why the two modules differ only in the definition of Eq. 2 and in the corresponding interpolation matrices $\mathbf{G}_x$ and $\mathbf{G}_y$, with $g_x = \mathbf{G}_x \bar{\mathbf{u}}$ and $g_y = \mathbf{G}_y \bar{\mathbf{u}}$. The additional term $-v/r$ of the Sanders-Koiter operator $\mathbf{G}_y$ is the same one highlighted in the geometric stiffness matrix of the linear buckling chapter. The constitutive relation is the usual laminated-shell one, collectively denoted $\mathbf{C}$, relating $\{\mathbf{N}, \mathbf{M}\}$ to $\{\boldsymbol{\varepsilon}, \boldsymbol{\kappa}\}$ through the ABD matrix.

2.1 State and rates are independent fields

The pre-buckling state is a curve $\hat{\mathbf{u}}(\lambda)$. The asymptotic expansion needs, besides the state itself, its first and second derivatives with respect to λ , written $\hat{\mathbf{u}}_{,\lambda}$ and $\hat{\mathbf{u}}_{,\lambda\lambda}$. Differentiating Eq. 1 along the path, and abbreviating $\hat{g}_\alpha = \mathbf{G}_\alpha \hat{\mathbf{u}}$, $g_{\alpha,\lambda} = \mathbf{G}_\alpha \hat{\mathbf{u}}_{,\lambda}$, $g_{\alpha,\lambda\lambda} = \mathbf{G}_\alpha \hat{\mathbf{u}}_{,\lambda\lambda}$, with $\alpha \in \{x, y\}$:

$$\boldsymbol{\varepsilon}_{,\lambda} = \mathbf{B}_m \hat{\mathbf{u}}_{,\lambda} + \begin{Bmatrix} \hat{g}_x g_{x,\lambda} \\ \hat{g}_y g_{y,\lambda} \\ \hat{g}_x g_{y,\lambda} + \hat{g}_y g_{x,\lambda} \end{Bmatrix}, \quad \boldsymbol{\kappa}_{,\lambda} = \mathbf{B}_b \hat{\mathbf{u}}_{,\lambda} \quad (3)$$

$$\boldsymbol{\varepsilon}_{,\lambda\lambda} = \mathbf{B}_m \hat{\mathbf{u}}_{,\lambda\lambda} + \begin{Bmatrix} g_{x,\lambda}^2 + \hat{g}_x g_{x,\lambda\lambda} \\ g_{y,\lambda}^2 + \hat{g}_y g_{y,\lambda\lambda} \\ 2g_{x,\lambda} g_{y,\lambda} + \hat{g}_x g_{y,\lambda\lambda} + \hat{g}_y g_{x,\lambda\lambda} \end{Bmatrix}, \quad \boldsymbol{\kappa}_{,\lambda\lambda} = \mathbf{B}_b \hat{\mathbf{u}}_{,\lambda\lambda} \quad (4)$$

and the derivatives with respect to the degrees-of-freedom:

$$\frac{\partial \boldsymbol{\varepsilon}}{\partial \bar{\mathbf{u}}} = \mathbf{B}_m + \begin{Bmatrix} \hat{g}_x \mathbf{G}_x \\ \hat{g}_y \mathbf{G}_y \\ \hat{g}_x \mathbf{G}_y + \hat{g}_y \mathbf{G}_x \end{Bmatrix}, \quad \frac{\partial^2 \boldsymbol{\varepsilon}}{\partial \lambda \partial \bar{\mathbf{u}}} = \begin{Bmatrix} g_{x,\lambda} \mathbf{G}_x \\ g_{y,\lambda} \mathbf{G}_y \\ g_{x,\lambda} \mathbf{G}_y + g_{y,\lambda} \mathbf{G}_x \end{Bmatrix} \quad (5)$$

If one assumes the pre-buckling path to be linear in the load parameter, $\hat{\mathbf{u}}(\lambda) = \lambda \hat{\mathbf{u}}(1)$, then $\hat{\mathbf{u}}_{,\lambda} = \hat{\mathbf{u}}(1)$, $\hat{\mathbf{u}}_{,\lambda\lambda} = \mathbf{0}$ and $\hat{g}_\alpha = \lambda g_{\alpha,\lambda}$, so Eqs. 3 to 5 collapse onto expressions containing λ explicitly. That assumption is precisely what removes the non-linear pre-buckling behaviour from the expansion, and it is not made here: state, first rate and second rate are carried as three independent nodal fields.

3 The tangent stiffness matrix

3.1 Definition and decomposition

The stiffness matrix used throughout is the derivative of the internal force vector with respect to the degrees-of-freedom. Differentiating the residual $\mathbf{R} = \mathbf{F}_{int} - \mathbf{F}_{ext}$, and with no follower forces, so that $\partial \mathbf{F}_{ext} / \partial \bar{\mathbf{u}} = \mathbf{0}$:

$$\mathbf{R}' = \frac{\partial \mathbf{R}}{\partial \bar{\mathbf{u}}} = \frac{\partial \mathbf{F}_{int}}{\partial \bar{\mathbf{u}}} = \mathbf{K}_T \quad (6)$$

Differentiating the internal force vector produces one contribution from the change of the stresses and another from the change of the differentiation operator itself:

$$\mathbf{K}_T = \underbrace{\int_{\Omega} (\mathbf{B}_L + \mathbf{B}_{NL})^\top \mathbf{C} (\mathbf{B}_L + \mathbf{B}_{NL}) d\Omega}_{\mathbf{K}_C} + \underbrace{\int_{\Omega} \frac{\partial (\mathbf{B}_L + \mathbf{B}_{NL})^\top}{\partial \bar{\mathbf{u}}} \boldsymbol{\sigma} d\Omega}_{\mathbf{K}_G} = \mathbf{K}_C + \mathbf{K}_G \quad (7)$$

For the elements used here the constitutive contribution is split further into the constant part and the part that depends on the displacement state:

$$\mathbf{K}_C(\bar{\mathbf{u}}) = \mathbf{K}_0 + \mathbf{K}_C^{NL}(\bar{\mathbf{u}}), \quad \mathbf{K}_0 = \int_{\Omega} \mathbf{B}_L^\top \mathbf{C} \mathbf{B}_L d\Omega \quad (8)$$

which matches the three assembly routines of the element library, `update_KC0`, `update_KCNL` and `update_KG`, so that $\mathbf{K}_T = \mathbf{K}_0 + \mathbf{K}_C^{NL} + \mathbf{K}_G$. The internal force vector is assembled separately, by `update_fint`.

3.2 Verifying consistency

Nothing in the assembly guarantees that the three matrices add up to the derivative of the separately assembled $\mathbf{F}_{int}$; that has to be verified. The test is a directional Taylor expansion. For an arbitrary direction $\mathbf{d}$ and a step ϵ :

$$e(\epsilon) = \frac{\|\mathbf{F}_{int}(\bar{\mathbf{u}} + \epsilon \mathbf{d}) - \mathbf{F}_{int}(\bar{\mathbf{u}}) - \epsilon \mathbf{K}_T \mathbf{d}\|}{\|\epsilon \mathbf{K}_T \mathbf{d}\|} \quad (9)$$

The numerator is $O(\epsilon^2)$ when $\mathbf{K}_T$ is the derivative, so $e(\epsilon)$ must fall proportionally to ϵ , dividing by ten at every decade, until round-off takes over. A plateau means the matrix is not the derivative. Measured on a single element with a displacement state of the order of the thickness:

ϵ	before <code>bfscylinder</code> 0.6.0	from 0.6.0 onwards
10^{-2}	6.186×10^{-2}	9.466×10^{-4}
10^{-3}	6.157×10^{-2}	9.470×10^{-5}
10^{-4}	6.154×10^{-2}	9.470×10^{-6}
10^{-5}	6.154×10^{-2}	9.470×10^{-7}
10^{-6}	6.154×10^{-2}	9.450×10^{-8}

A second test identifies where a deficiency sits. Taking the true directional derivative $\mathbf{J}\mathbf{d}$ from a central difference and fitting the coefficients of the two state-dependent matrices by least squares,

$$\min_{c_1, c_2} \|\mathbf{K}_0 \mathbf{d} + c_1 \mathbf{K}_C^{NL} \mathbf{d} + c_2 \mathbf{K}_G \mathbf{d} - \mathbf{J}\mathbf{d}\| \quad (10)$$

returns $c_1 = c_2 = 1.0000$ with a residual of 7.6×10^{-11} from 0.6.0 onwards. Before 0.6.0 it returned $c_1 = 1.5122$ and $c_2 = 1.2876$ with a residual of 3.8×10^{-2} , against 6.0×10^{-2} for the assembled combination: no recombination of the three matrices reproduced the derivative, which identified the defect as a missing term rather than a mis-scaled one. Both kinematics are covered, the Sanders-Koiter element reaching 1.2×10^{-10} .

3.3 Why the consistency matters twice

An inconsistent tangent stiffness matrix has two distinct consequences here, and only the first is the familiar one.

In the Newton-Raphson solution of Section 4 the tangent controls the rate of convergence only, the converged state being fixed exclusively by the residual, that is by the accuracy with which $\mathbf{F}_{int}$ and $\mathbf{F}_{ext}$ are computed. An inconsistent tangent therefore degrades the iteration from quadratic to linear, with a contraction factor that grows as the tangent is approached, but it does not by itself move the equilibrium state. On the coarse meshes of Section 11 the contraction was about 0.3 per iteration, which still converges; on a mesh with $n_y = 60$ it reached 0.95, exhausting the iteration budget at every load step and leaving the load stepping stalled near $\lambda_b/\lambda_c = 0.94$.

The second consequence has no such protection. The matrix $\mathbf{K}_C = \mathbf{K}_0 + \mathbf{K}_C^{NL}$ does not only drive the iteration: it enters the eigenvalue problem of Eq. 15 and the expansion operator of Eq. 23 directly, where it is not a preconditioner but part of the statement of the problem. An error in $\mathbf{K}_C^{NL}$ is therefore inherited by λ_c , by the buckling mode and by every post-buckling coefficient. Correcting it moved the buckling loads of Section 11 by 9 to 11 per cent, onto the reference solutions, and reversed the sign of b_{1111} .

4 The axisymmetric pre-buckling solver

4.1 The fundamental path is axisymmetric

The fundamental path of a perfect cylinder under uniform axial compression is axisymmetric, and every buckling mode that makes $\mathbf{K}_T$ singular is not. Solving the pre-buckling problem in the axisymmetric subspace therefore keeps the iteration clear of the entire near-critical cluster, and not merely of the few modes that happen to have been computed. This is also what the reference solutions do, AXBIF and ANILISA both solving an axisymmetric pre-buckling problem.

4.2 Cyclic symmetry of the mesh

The mesh has n_y elements around the circumference, and the element degrees-of-freedom are expressed in the local shell frame, which turns with y . Shifting the circumferential node index cyclically by one element is therefore a symmetry of every operator assembled on the mesh: if $\mathbf{P}$ denotes that shift, then $\mathbf{P}^\top \mathbf{K} \mathbf{P} = \mathbf{K}$ for $\mathbf{K} \in \{\mathbf{K}_0, \mathbf{K}_C^{NL}, \mathbf{K}_G\}$ whenever the state they are evaluated at is itself invariant.

Two consequences are used. The first is that the average over the orbit of $\mathbf{P}$ is the orthogonal projector onto the invariant subspace, used to remove the round-off asymmetry of the starting state. The second, used in Section 6, is that $\mathbf{P}$ maps a buckling mode onto another buckling mode carrying the same multiplier.

4.3 The reduced problem

Let $\mathbf{B}_{axi}$ collect one column per axial station and per degree-of-freedom, carrying one at that degree-of-freedom of every node of the station,

$$[\mathbf{B}_{axi}]_{(i n_y + j) n_{DOF} + d, i n_{DOF} + e} = \delta_{de}, \quad \begin{array}{l} i = 0 \dots n_x - 1 \\ j = 0 \dots n_y - 1 \\ d, e = 0 \dots n_{DOF} - 1 \end{array} \quad (11)$$

so that an axisymmetric field is $\mathbf{B}_{axi} \mathbf{a}$, the columns are mutually orthogonal, and $\mathbf{B}_{axi}^\top \mathbf{B}_{axi} = n_y \mathbf{I}$.

The correction is then obtained by a Bubnov-Galerkin projection [3]: it is sought *in* the subspace spanned by the columns of $\mathbf{B}_{axi}$, and the residual of the tangent problem is required to be orthogonal *to that same subspace*,

$$\delta\bar{\mathbf{u}} = \mathbf{B}_{axi}\mathbf{a} \quad (\text{trial}), \quad \mathbf{B}_{axi}^\top (\mathbf{K}_T \mathbf{B}_{axi}\mathbf{a} + \mathbf{R}) = \mathbf{0} \quad (\text{test}) \quad (12)$$

Trial and test spaces coincide, which is what makes the projection Bubnov-Galerkin rather than Petrov-Galerkin, and which preserves the symmetry of $\mathbf{K}_T$ in the reduced operator. Written out, Eq. 12 is

$$\left(\mathbf{B}_{axi}^\top \mathbf{K}_T \mathbf{B}_{axi} \right) \mathbf{a} = -\mathbf{B}_{axi}^\top \mathbf{R}, \quad \delta\bar{\mathbf{u}} = \mathbf{B}_{axi}\mathbf{a} \quad (13)$$

This is the reduced-basis technique for non-linear structural analysis of Noor and Peters [5], used here with a basis known *a priori* from the symmetry of the problem instead of one accumulated from previous solutions. The cyclic shift $\mathbf{P}$ commutes with $\mathbf{K}_T$, and the column space of $\mathbf{B}_{axi}$ is exactly the subspace invariant under it; building the basis from the symmetry group in this way is the standard treatment of rotationally periodic structures [7]. What is specific to the present use is that only the zeroth circumferential harmonic is retained, the remaining $n_y - 1$ harmonics being precisely where the buckling modes live.

Three consequences follow.

1. **Size.** The reduced system has $n_x \times n_{DOF}$ unknowns against the full count, 250 against 15000 on the $n_y = 60$ mesh, and is small enough to be solved as a dense system.
2. **No deflation.** The whole near-critical cluster lies outside the subspace by construction, so the buckling modes never have to be deflated out of the correction.
3. **The reduced solution is exact.** Because $\mathbf{K}_T$ and $\mathbf{R}$ commute with $\mathbf{P}$, the residual evaluated at an axisymmetric state is itself axisymmetric and therefore has no component outside the subspace. Hence $\mathbf{B}_{axi}^\top \mathbf{R} = \mathbf{0}$ implies $\mathbf{R} = \mathbf{0}$ on the free degrees-of-freedom, and the Galerkin condition of Eq. 12 is *equivalent* to the full equilibrium equations rather than a projection of them. The converged state solves the full problem exactly. This is what separates a symmetry-adapted basis from a general reduced basis, for which Eq. 12 leaves a residual outside the subspace that is never driven to zero.

A reduced coordinate is treated as known as soon as one of the nodes it spans is constrained. This matters for the single-node constraint that suppresses the axial rigid body translation, which is the one part of the boundary conditions that is not rotationally symmetric: in the reduced problem it fixes the whole axial station, which is what an axisymmetric field requires of it anyway. The same asymmetry is handled in Section 6 by subtracting an axial rigid body translation, which carries no strain and is annihilated by both stiffness matrices.

4.4 Conditioning of the reduced matrix

The reduced coordinates place nodal displacements next to nodal derivatives, whose stiffness entries differ by many orders of magnitude, and the free part of $\mathbf{K}_r = \mathbf{B}_{axi}^\top \mathbf{K}_T \mathbf{B}_{axi}$ reaches a condition number of 4×10^{12} on the $n_y = 60$ mesh. The sparse solver used for the full systems equilibrates internally; the dense solver used on the reduced one does not, so the symmetric (Jacobi) equilibration is applied explicitly.

Let $\mathbf{D} = \text{diag}(|K_{r,kk}|^{-1/2})$. Then $\mathbf{K}_r \mathbf{a} = \mathbf{r}$ is solved in the form

$$(\mathbf{D}\mathbf{K}_r\mathbf{D})(\mathbf{D}^{-1}\mathbf{a}) = \mathbf{D}\mathbf{r}, \quad \text{so} \quad \mathbf{a} = \mathbf{D} \left[(\mathbf{D}\mathbf{K}_r\mathbf{D})^{-1} \mathbf{D}\mathbf{r} \right] \quad (14)$$

which is what the code implements: it forms $\mathbf{D}\mathbf{K}_r\mathbf{D}$ and $\mathbf{D}\mathbf{r}$, solves the scaled system, and multiplies the result by $\mathbf{D}$ once more. The scaled matrix has unit diagonal in absolute value.

For a symmetric positive definite matrix, equilibrating the diagonal in this way is within a factor of n of the optimal symmetric scaling [8], n being the order of the matrix; the reduced tangent is positive definite everywhere the load stepping actually solves it, which is on the near side of the bifurcation point. The measured condition number falls from 4×10^{12} to 3×10^2 .

Two points are worth making explicit. Eq. 14 is an exact rewriting of the same linear system, not a preconditioner and not an approximation: it changes the floating-point path taken, never the solution sought. And on the meshes tried it changes nothing measurable — on the $n_y = 60$ case every iterate is unchanged to ten digits, the graded matrix being solved accurately by partial pivoting alone. It is kept as a guard against worse-conditioned meshes, not as a correction to these.

5 Locating the expansion point

5.1 Why the load level matters

Koiter’s expansion is asymptotic about the bifurcation point. Sun §2.2 requires the pre-buckling state to satisfy $\lambda_b/\lambda_c \approx 0.995$. If instead the pre-buckling problem is solved once at the reference load and the eigenvalue problem is posed about that state, then for the shells considered here λ_b/λ_c is of the order of 0.1. At that load level the radial pre-buckling deflection is one to three per cent of a wall thickness, the boundary layer produced by the edge restraint has not developed, and $\|\mathbf{K}_C^{NL}(\hat{\mathbf{u}})\|/\|\mathbf{K}_0\| \sim 2 \times 10^{-4}$; the eigenvalue analysis is then a linear buckling analysis in all but name and the whole computation returns the *membrane* pre-buckling answer.

5.2 Iterative eigenvalue algorithm

The expansion point is located by the incremental-iterative scheme of Sun Eqs. (44) to (46). At the current load level λ_b the pre-buckling state $\hat{\mathbf{u}}$ is known and the eigenvalue problem

$$[\mathbf{K}_C(\hat{\mathbf{u}}) + \mu \mathbf{K}_G(\hat{\mathbf{u}})] \delta \bar{\mathbf{u}} = \mathbf{0} \quad (15)$$

gives the multiplier μ of the *current stress state*, so that $\lambda_c = \lambda_b \mu$. This is the eigenvalue problem of the linear buckling chapter with $\mathbf{K}_0$ replaced by the constitutive part of the tangent at the current state.

5.3 Advancing the load level

Sun Eq. (44) advances the load level by a fixed fraction of the distance to the bifurcation load:

$$\lambda_b \leftarrow \lambda_b + \eta (\lambda_c - \lambda_b) \quad (16)$$

with $\eta = 0.8$ on the first step and $\eta = 0.5$ afterwards. The difficulty is that λ_c is not fixed: it is re-evaluated at the new state and it falls as λ_b rises. Writing $s = d\lambda_c/d\lambda_b$ and linearising, the new distance to the bifurcation point is

$$\lambda'_b - \lambda'_c = (\lambda_c - \lambda_b) [\eta(1 - s) - 1] \quad (17)$$

so the step lands *past* the bifurcation point whenever

$$\eta > \frac{1}{1 - s} \quad (18)$$

The sensitivity s steepens as the bifurcation point is approached and as the mesh is refined. On the $n_y = 40$ meshes of Section 11 it reaches -0.25 and -0.72 , for which Eq. 18 permits $\eta = 0.80$ and $\eta = 0.58$, both above the fixed 0.5, and no overshoot occurs. On the $n_y = 60$ mesh it reaches -1.09 , permitting only $\eta = 0.478$, and the fixed $\eta = 0.5$ overshoots to $\lambda_b/\lambda_c = 1.0013$.

The implementation therefore estimates s from the two previous load steps and caps the step at a fraction f of the permissible value:

$$\eta \leftarrow \min \left(\eta, \frac{f}{1-s} \right), \quad s = \frac{\lambda_c - \lambda_c^{prev}}{\lambda_b - \lambda_b^{prev}}, \quad f = 0.7 \quad (19)$$

Substituting Eq. 19 into Eq. 17 gives $\lambda'_b - \lambda'_c = -(1-f)(\lambda_c - \lambda_b)$ whatever the value of s , so the cap does not merely prevent the overshoot: it turns Eq. 16 into a secant iteration on $\lambda_c(\lambda_b) - \lambda_b$, which divides the remaining distance by a fixed factor at every step.

The fraction is 0.7 rather than something closer to unity because s is estimated backwards and lags while it steepens. On the $n_y = 60$ mesh the offending step is taken with $s = -0.66$ measured over the previous interval, which permits $\eta = 0.54$ and therefore does not bind on the fixed $\eta = 0.5$, while the step itself turns out to have $s = -1.09$. A margin of about a quarter on $1-s$ absorbs that lag, and with $f = 0.7$ the same mesh lands at $\lambda_b/\lambda_c = 0.9965$ instead of 1.0013, in the same number of load steps.

5.4 Accepting the state

Convergence is declared by Sun Eq. (45). That criterion measures the distance to the bifurcation point from below only, so that once λ_b passes λ_c its left-hand side turns negative and it accepts the state whatever the overshoot. The distance is therefore taken in absolute value:

$$\frac{|\lambda_c - \lambda_b|}{\lambda_c} \leq \varepsilon_1, \quad \varepsilon_1 = 0.005 \quad (20)$$

If the load level lands past the bifurcation point by more than ε_1 , the load step is taken again from the previous state with η halved. The fundamental path is still computable there, the pre-buckling solve being restricted to the axisymmetric subspace, but the expansion is meant to be made on the near side of it.

At convergence $\mu \simeq 1$ to within ε_1 , which is what makes the operators of Eq. 15 usable directly in the expansion.

5.5 Rates of the pre-buckling state

The first rate follows from Sun Eq. (29), using the tangent stiffness at the converged pre-buckling state and solved in the reduced basis of Eq. 13:

$$\mathbf{K}_T \hat{\mathbf{u}}_{,\lambda} = \mathbf{F}_{ref} \quad (21)$$

The second rate follows from Sun Eq. (35) as a backward difference against the load step preceding the converged one:

$$\hat{\mathbf{u}}_{,\lambda\lambda} = \frac{\hat{\mathbf{u}}_{,\lambda} - \hat{\mathbf{u}}_{,\lambda}^{prev}}{\lambda_b - \lambda_b^{prev}} \quad (22)$$

Reusing a step the load stepping has already converged on costs nothing and, unlike a step solved ahead of λ_b , cannot fall on the far side of the bifurcation point. The corresponding stress rates, Sun Eqs. (30) and (34), follow from Eqs. 3 and 4 through the constitutive matrix.

6 Buckling modes of a cylinder

6.1 The modes come in degenerate pairs

For a circular cylinder the buckling modes occur in degenerate pairs, one for each sign of the circumferential wave number, so $\dim \ker \geq 2$ even when a single Koiter mode is retained, and

every rotation of a pair inside its own eigenspace is again a pair of buckling modes. Which member of it the eigen solver returns is decided by round-off, and changes with the BLAS implementation.

That would not matter if the post-buckling coefficients were invariant under that rotation, and they are not. The second-order field of a mode with n circumferential waves carries the harmonic $2n$, which a mesh sized for the buckling mode itself barely resolves. Measured on the case of Section 11 with $n_y = 40$ and $n = 10$, rotating the critical pair by 30 degrees moves b_{1111} from -0.230556 to -0.192752 , sixteen per cent, with the pre-buckling state and the buckling load unchanged to eleven digits. Before the tangent of Section 3 was made consistent the same rotation moved it from -0.0008 to $+0.2744$, sign included, so most of that dependence was the tangent and what remains is the mesh.

6.2 Fixing the member

Each pair is rotated to the member whose crest falls on the $y = 0$ generator, which is a property of the mesh and not of the arithmetic. Of the two members of a pair one has a crest on that generator and the other a node, so the dominant eigenvector of the 2×2 Gram matrix of their traces there is well separated, and the sign is fixed by requiring the crest to point outwards.

The partner of a mode need not be among the ones the eigen solver returned. A Krylov method builds its subspace from a single starting vector, so it finds one direction of a degenerate eigenspace and then moves on to the next multiplier: asking for the two lowest multipliers of the shell of Section 11 returns one member of the critical pair and one member of the next pair, 3.3×10^{-4} away. A missing partner is recovered through the cyclic shift $\mathbf{P}$ of Section 4, which maps a mode onto another mode of the same pair, after removal of the axial rigid body translation that restores the single-node constraint. Truncating the offending degree-of-freedom instead takes the mode out of its eigenspace and, the operator of the second-order field being singular along it, the error is then amplified without bound.

This makes the coefficients reproducible: 9×10^{-9} between a single threaded and a multi threaded run, against a sign difference between operating systems before. It does not make them mesh converged, and only a mesh that resolves the harmonic $2n$ would.

6.3 How many modes the expansion actually needs

Fixing the member makes a single-mode result reproducible. It does not make it complete, and the measured spectra say by how much it falls short. Twelve multipliers were computed about the converged pre-buckling state of each reference shell, $n_y = 40$; every one of them came out as a degenerate pair, to eleven digits, so the cluster is counted in pairs below.

	pairs within			gap after the 1% cluster
	0.5%	1%	2%	
Sun §3.1, NLprebuck=False	1	3	5	0.28%
Sun §3.1, NLprebuck=True	3	4	4	2.29%
Arbocz, NLprebuck=False	1	4	6	0.26%
Arbocz, NLprebuck=True	2	3	4	0.99%

Two readings of this table matter.

The first is that the single-mode expansion retains *one* eigenvector out of six or eight. Its very first omission is the degenerate partner of the critical mode, which Section 6 goes to some trouble to construct and then discards. That alone bounds what the single-mode coefficient can be expected to reproduce, and it is consistent with the residual five per cent between b_{1111} and the ANILISA value of Section 11 surviving every mesh refinement tried. Closing the 1%

cluster requires `koiter_num_modes` = 6 for the Arbocz shell and 8 for the Sun shell; closing only the 0.5% cluster requires 4 and 6. In pairs, which is the natural unit here, that is three to four pairs.

The second is that the non-linear pre-buckling state does not merely move the modes, it *separates* them. With `NLprebuck=False` the cluster has no boundary worth the name, the next pair sitting 0.26 to 0.28 per cent above the last one retained, so any truncation is arbitrary. With `NLprebuck=True` a genuine spectral gap opens, 0.99 and 2.29 per cent, an order of magnitude wider than the spacing inside the cluster. Only then is “the cluster” a well-defined set, and only then is a multi-mode truncation of it well posed. The pre-buckling treatment of this document is therefore a prerequisite for multi-mode Koiter, not an alternative to it.

Two pieces of the implementation stand between this and a working multi-mode analysis. The symmetrisation of ϕ_{30} over distinct mode indices is established only for coinciding indices (Section 13), and the row border $\mathbf{W}$ of the bordered system (Section 8) carries one condition per retained Koiter mode while the null space of ϕ_2 has as many dimensions as the cluster has members, so the two counts have to be brought into agreement before the extra modes can be retained rather than deflated.

7 Consistency of the operators

The operator appearing in the asymptotic expansion,

$$\phi_2 = \mathbf{K}_C(\hat{\mathbf{u}}) + \mu \mathbf{K}_G(\hat{\mathbf{u}}) \quad (23)$$

must be *the same* operator whose null vector is the buckling mode, that is, exactly the operator of the eigenvalue problem of Eq. 15. Building ϕ_2 from a differently evaluated $\mathbf{K}_C^{NL}$ than the one used there, for instance from $\mathbf{K}_C^{NL}(\lambda_c \hat{\mathbf{u}})$ while the eigenvalue problem used $\mathbf{K}_C^{NL}(\hat{\mathbf{u}})$, leaves ϕ_2 non-singular along $\delta \bar{\mathbf{u}}$ and corrupts every quantity derived from it. The severity is easy to measure: with the inconsistent pairing the expansion operator does not annihilate the buckling mode at all, the relative residual rising from 10^{-10} to 4.6. Since $\lambda_b \simeq \lambda_c$ after the iterative algorithm has converged, $\mu \simeq 1$ and Eq. 23 is simply the tangent operator at the critical state.

The discrete energy derivatives assembled element by element are the building blocks of the expansion:

symbol in the code	meaning
<code>phi2</code>	$\phi_2 = \partial^2 \phi / \partial \bar{\mathbf{u}}^2$, the tangent operator at the critical state
<code>phi3_ab[(i,j)]</code>	ϕ_3 contracted with modes i, j , one free index
<code>phi4[(i,j,k,l)]</code>	ϕ_4 fully contracted
<code>phi20_a[i]</code>	$\partial \phi_2 / \partial \lambda$ contracted with mode i , one free index
<code>phi30_ab[(i,j)]</code>	$\partial \phi_3 / \partial \lambda$ contracted with modes i, j , one free index
<code>phi200_ab[(i,j)]</code>	$\partial^2 \phi_2 / \partial \lambda^2$ fully contracted

All of them are built from Eqs. 3 to 5 together with the constant quadratic form $\partial^2 \varepsilon / \partial \bar{\mathbf{u}} \partial \bar{\mathbf{u}}$, which for Eq. 1 is $\mathbf{G}_x^\top \mathbf{G}_x$, $\mathbf{G}_y^\top \mathbf{G}_y$ and $\mathbf{G}_x^\top \mathbf{G}_y + \mathbf{G}_y^\top \mathbf{G}_x$ in the three strain components.

8 Second-order fields and the orthogonality condition

8.1 The singular system

The second-order fields $\bar{\mathbf{u}}_{ij}$ solve

$$\phi_2 \bar{\mathbf{u}}_{ij} = \mathbf{g}_{ij}, \quad \mathbf{g}_{ij} = -\frac{1}{2} \phi_{3,ij} - \frac{1}{m} \sum_k a_{kij} \lambda_k \phi_{20,k} \quad (24)$$

with m the number of Koiter modes. By construction ϕ_2 is singular, its null space being spanned by the buckling modes. Consequently Eq. 24 determines $\bar{\mathbf{u}}_{ij}$ only up to an arbitrary element of that null space, and a plain sparse solve followed by a projection is not a well-posed procedure. Two things are therefore needed: a solver that copes with the singularity, and the correct condition that fixes the null-space component.

8.2 The condition to be imposed

The orthogonality condition of Koiter's theory is Sun Eq. (18), whose finite element form is Sun Eq. (33):

$$\delta \bar{\mathbf{u}}_k^\top [\mathbf{K}_D(\hat{\mathbf{u}}, \hat{\mathbf{u}}_{,\lambda}) + \mathbf{K}_G(\boldsymbol{\sigma}_{,\lambda})] \bar{\mathbf{u}}_{ij} + \frac{1}{2} \delta \bar{\mathbf{u}}_k^\top \mathbf{B}_{NL}^\top (\delta \bar{\mathbf{u}}_k) \mathbf{C} \mathbf{B}_{NL} (\delta \bar{\mathbf{u}}_k) \hat{\mathbf{u}}_{,\lambda} = 0 \quad (25)$$

Two observations turn this into something directly computable. The bracket in the first term is the load-parameter derivative of the tangent operator, $\partial \phi_2 / \partial \lambda$, so the first term is exactly the contraction stored as `phi20_a[k]`. The second term is a constant, which follows from ϕ_{30} without any new element integral: $\phi_{30,ij}$ is assembled as one half of the quadrature weight times the sum of six contractions which, when all mode indices coincide, are equal to one another by the symmetry of the constitutive matrix and of the quadratic form. Hence

$$c_k = \frac{\phi_{30,ij} \cdot \delta \bar{\mathbf{u}}_k}{6} \quad (26)$$

and the condition to impose is

$$\phi_{20,k} \cdot \bar{\mathbf{u}}_{ij} = -c_k \quad (27)$$

It is weighted by the stiffness matrices, through ϕ_{20} , and inhomogeneous, through c_k . A Gram-Schmidt projection, or equivalently a symmetric Euclidean border, imposes neither of these properties and is the procedure that Eq. 27 replaces.

8.3 The bordered system

Both requirements are met by a *non-symmetric* bordered system:

$$\begin{bmatrix} \phi_2 & \mathbf{N}_{sp} \\ \mathbf{W}^\top & \mathbf{0} \end{bmatrix} \begin{Bmatrix} \bar{\mathbf{u}}_{ij} \\ \boldsymbol{\alpha} \end{Bmatrix} = \begin{Bmatrix} \mathbf{g}_{ij} \\ -\mathbf{c} \end{Bmatrix} \quad (28)$$

in which the roles of the two borders are distinct. The *column* border $\mathbf{N}_{sp}$ spans the null space of ϕ_2 and is what makes the system non-singular; the *row* border $\mathbf{W}$ carries the constraints, one column per Koiter mode, $\mathbf{W}[:,k] = \phi_{20,k}$, with the corresponding right-hand side $-c_k$ from Eq. 26. The system is non-singular provided $\mathbf{W}^\top \mathbf{N}_{sp}$ is, and its leading entry $\phi_{20,k} \cdot \delta \bar{\mathbf{u}}_k$ is exactly the quantity appearing in the denominator of the post-buckling coefficients, hence non-zero by construction.

Because the critical eigenspace is degenerate, $\mathbf{N}_{sp}$ must span the *whole* near-null space and not only the retained Koiter modes: every eigenvector whose multiplier equals the critical one within a relative tolerance is included, and the set is orthonormalised with a rank-revealing factorisation. Equation 25 supplies one condition per Koiter mode, which is fewer than the dimension of the null space; the degenerate partners lie outside single-mode theory, and the remaining rows of $\mathbf{W}$ keep the Euclidean condition. This is a choice, not a result; the consistent treatment of a degenerate critical mode is the multi-mode Koiter expansion.

8.4 What the weighted condition does, and does not, change

Condition 27 is the correct one and replaces a Gram-Schmidt projection that was neither weighted nor inhomogeneous. For the cases considered here, however, it changes almost nothing. Measured before the tangent was made consistent, running the same analysis under the two conditions with everything else identical moved the second-order field by 6.5×10^{-7} relative, so the two conditions are numerically near-coincident for this problem: the null-space component of $\bar{\mathbf{u}}_{ij}$ on which they disagree is small to begin with.

A second reason reinforces this. Condition 27 fixes only the component of $\bar{\mathbf{u}}_{ij}$ along $\ker \phi_2$, and b_{ijkl} sees $\bar{\mathbf{u}}_{ij}$ through the contractions $\phi_{3,ij} \cdot \bar{\mathbf{u}}_{kl}$; but $\phi_{3,ij}$ contracted with a buckling mode is precisely the numerator of a_{ijk} in Eq. 29. For the symmetric bifurcation of a cylinder under axial compression a_{ijk} vanishes, measured values being of order 10^{-5} against a b_{ijkl} of order unity, so $\phi_{3,ij}$ is nearly orthogonal to the null space. For an asymmetric bifurcation, with $a_{ijk} \neq 0$, the condition would carry more weight.

9 Post-buckling coefficients

With the second-order fields available, the coefficients are

$$a_{ijk} = -\frac{1}{2\lambda_i} \frac{\phi_{3,ij} \cdot \delta \bar{\mathbf{u}}_k}{\phi_{20,i} \cdot \delta \bar{\mathbf{u}}_i} \quad (29)$$

$$b_{ijkl} = -\frac{1}{6\lambda_i(\phi_{20,i} \cdot \delta \bar{\mathbf{u}}_i)} \left[\phi_{4,ijkl} + 3\phi_{3,ij} \cdot \bar{\mathbf{u}}_{kl} + 3\phi_{3,il} \cdot \bar{\mathbf{u}}_{jk} \right. \\ \left. + \lambda_i(a_{iij}\phi_{30,ik} \cdot \delta \bar{\mathbf{u}}_l + a_{ijk}\phi_{30,il} \cdot \delta \bar{\mathbf{u}}_i + a_{ikl}\phi_{30,ii} \cdot \delta \bar{\mathbf{u}}_j) \right. \\ \left. + \phi_{200,ii}\lambda_i^2(a_{iij}a_{ikl} + a_{ijk}a_{ili} + a_{ikl}a_{iij}) \right] \quad (30)$$

The buckling modes are normalised so that the largest translational amplitude equals the shell thickness, which is the normalisation used by the reference solutions.

10 Algorithm

Algorithm 1 Koiter analysis about a non-linear pre-buckling state

```

1: solve  $\mathbf{K}_0 \bar{\mathbf{u}} = \mathbf{F}_{ref}$ ; project onto the axisymmetric subspace;  $\lambda_b \leftarrow 1$ 
2: for  $i = 1, \dots, i_{\max}$  do
3:   assemble  $\mathbf{K}_C(\hat{\mathbf{u}})$ ,  $\mathbf{K}_G(\hat{\mathbf{u}})$ 
4:   solve Eq. 15  $\rightarrow \mu$ ,  $\delta \bar{\mathbf{u}}$ ; canonicalise the degenerate pairs (Section 6);  $\lambda_c \leftarrow \lambda_b \mu$ 
5:   if Eq. 20 holds then break
6:   if  $\lambda_b > \lambda_c$  then restore the previous step, halve the cap on  $\eta$ , continue
7:    $\eta \leftarrow 0.8$  if  $i = 1$  else 0.5, capped by Eq. 19
8:   repeat
9:      $\lambda_b^{new} \leftarrow \lambda_b + \eta(\lambda_c - \lambda_b)$ 
10:    predictor  $\hat{\mathbf{u}} + \hat{\mathbf{u}}_{,\lambda}(\lambda_b^{new} - \lambda_b)$ ; Newton-Raphson in the reduced basis (Eq. 13)
11:    on failure  $\eta \leftarrow \eta/2$ 
12:  until converged or  $\eta$  exhausted
13:  store  $(\lambda_b, \lambda_c, \hat{\mathbf{u}}_{,\lambda})$  as the previous step; advance
14: end for
15:  $\hat{\mathbf{u}}_{,\lambda}$  from Eq. 21;  $\hat{\mathbf{u}}_{,\lambda\lambda}$  from Eq. 22
16: assemble  $\phi_2, \phi_3, \phi_4, \phi_{20}, \phi_{30}, \phi_{200}$ 
17:  $a_{ijk}$  from Eq. 29
18:  $\bar{\mathbf{u}}_{ij}$  from the bordered system Eq. 28 with the weighted condition Eq. 27
19:  $b_{ijkl}$  from Eq. 30

```

11 Effect on the results

The two verification cases are the composite cylinder of Sun §3.1 and the NASA shell AW-CYL-1-1 of Arbocz et al. [1]; they are the same shell, so the ANILISA n -search of the latter provides reference values for both. Buckling loads are normalised by $N_{c\ell} = E_{11} h^2 / (r \sqrt{3(1 - \nu_{12}^2)})$.

11.1 Buckling loads

	n	λ_c	reference
Arbocz, $n_y = 40$	10	0.331413	0.329163 (ANILISA, $n = 10$)
Arbocz, $n_y = 60$	11	0.330603	0.328594 (ANILISA, $n = 11$)
			0.327759 (STAGS-A, $n = 11$)
Sun, $n_y = 40$	10	0.346127	—
Sun, $n_y = 50$	10	0.347244	—

The two Arbocz meshes agree with the ANILISA entry for their own circumferential wave number to 0.7 and 0.6 per cent respectively, and the finer one with the Level-3 STAGS-A value to 0.9 per cent. The membrane pre-buckling regime, which is what the computation returns when the expansion point is left at the reference load, gives 0.365992 instead.

11.2 Post-buckling coefficients

	n	b_{1111}	reference
Arbocz, $n_y = 40$	10	-0.335975	—
Arbocz, $n_y = 60$	11	-0.356457	-0.37605 (ANILISA, $n = 11$) -0.3772 (Sun, Table 2)
Sun, $n_y = 40$	10	-0.230556	—
Sun, $n_y = 50$	10	-0.222844	—

The $n_y = 60$ value is within five per cent of the ANILISA coefficient for the same mode. A negative b means an imperfection sensitive shell, which is what this one is.

11.3 What the tangent correction changed

Before the tangent stiffness matrix was made consistent (Section 3.2) the same meshes returned $\lambda_c = 0.302887$ and $b_{1111} = +0.410664$ for the Arbocz case: the buckling load was nine per cent low and the post-buckling coefficient had the wrong sign. The sign was at that time attributed to a single-mode expansion about a degenerate critical mode being unable to determine it, which was incorrect. Two meshes that did not even select the same critical mode, $n = 10$ at $n_y = 40$ against $n = 7$ at $n_y = 50$ for the Sun case, now select $n = 10$ on both and agree to 0.3 per cent in the buckling load and 3.4 per cent in b_{1111} .

The Newton-Raphson behaviour changed with it. Each load step now takes a single iteration, with residuals falling 5.1×10^{-5} , 1.1×10^{-5} , 1.2×10^{-6} , 5.4×10^{-8} , 1.5×10^{-9} , and no step back-off anywhere. The $n_y = 60$ mesh, which previously exhausted the iteration budget at every load step, converges in four of them.

12 The variable-stiffness CTS model

The routines `fkoiter_cylinder_CTS_circum` of `koiter_cylinder_CTS.py` and `koiter_cylinder_CTS_sander` apply the whole of the preceding development to the continuous-tow-shearing (CTS) variable-stiffness cylinders of dos Santos and Castro, *Lightweight design of variable-stiffness cylinders with reduced imperfection sensitivity enabled by continuous tow shearing and machine learning*, Materials **15** (2022) 4117. The algorithm is taken literally from the constant-stiffness models, and a check in the test suite asserts that the shared regions of the two files are identical character for character, so only what is specific to the CTS model is documented here.

12.1 Why the axisymmetric reduction still applies

The parameterisation steers the tows along the *axial* direction only, $\theta = \theta(x)$, and the tow shearing thickens the laminate as $h_{tow}/\cos\theta$, again a function of x alone. The stiffness is therefore uniform around the circumference, the cyclic shift of Section 4 remains a symmetry of every assembled operator, and the fundamental path of the perfect cylinder under uniform axial compression is still axisymmetric. The Galerkin reduction of Eq. 13 carries over unchanged.

Two consequences are worth stating explicitly, because they are easy to confuse:

1. The reduction constrains the **pre-buckling state only**. The eigenvalue problem is assembled and solved on the full space, so non-axisymmetric buckling modes are obtained exactly as they are in the constant-stiffness models. What the reduction removes is the possibility of the pre-buckling iteration drifting off the fundamental path, not the modes.
2. The *premise* of the reduction — that every buckling mode making $\mathbf{K}_T$ singular is non-axisymmetric — is **not** satisfied everywhere in the CTS design space. See Section 12.5.

Unlike the constant-stiffness models, the axial station spacing is not uniform: it is built from the transition length $t = r_{CTS} \sin|\theta_{c2} - \theta_{c1}|$ and the plateau lengths c_1 and c_2 . An assertion checks that the stations `mesh_order` sorts the nodes into are the ones the axisymmetric basis is then built on, which a non-uniform x grid could otherwise violate silently.

12.2 Length scales that size the mesh

Three scales control the discretisation. With $r = 68$ mm, $L = 300$ mm and $h \simeq 0.8$ mm, the geometry of the cylinder used throughout the test suite, they are

scale	value	constrains
$\sqrt{rh}$	7.4 mm	—
edge boundary layer $1/\beta = \sqrt{rh} / [3(1 - \nu^2)]^{1/4}$	5.8 mm	n_x
buckling half-wave $\pi\sqrt{rh} / [12(1 - \nu^2)]^{1/4}$	12.8 mm	n_x
wavelength of the $2n$ harmonic, $\pi r/n$	21 mm at $n=10$	n_y

The axial mesh is set by the edge boundary layer. With `NLprebuck=True` the non-linear pre-buckling deformation caused by the edge restraint is concentrated within a few multiples of $1/\beta$ of each end, and it is precisely that deformation which the expansion is about; with `NLprebuck=False` the requirement is much weaker, the linear pre-buckling state being smooth. This is a demand the earlier implementation did not make, because expanding at the reference load left the boundary layer undeveloped.

The circumferential mesh is set by the *second-order field*, not by the buckling mode. A mode with n circumferential waves drives a second-order field carrying the harmonic $2n$ (Section 13), so a mesh sized to resolve the mode itself samples the field that determines b_{ijkl} at only about two nodes per wave. Resolving the $2n$ harmonic with four nodes per wave requires $n_y \simeq 8n$.

12.3 Measured convergence

The design below is representative of the parameterisation: $\theta_{c1} = 30^\circ$, $\theta_{c2} = 60^\circ$, $n = 3$, $c_{2,ratio} = 0.5$, $r_{CTS} = 50$ mm, giving $t = 25$ mm, $c_1 = 18.75$ mm and $c_2 = 25$ mm. Sanders kinematics, `idealistic_CTS=True`, `NLprebuck=False`, four integration points. The axial mesh is refined with the circumferential one through `nxt`, which keeps the element aspect ratio between 1.0 and 1.4.

n_y	n_x	DOF	Δx	$\Delta x \beta$	P_{cr} [N]	ΔP_{cr}
30	23	6 900	13.64 mm	2.37	40982.6	+6.65%
45	37	16 650	8.33 mm	1.45	38893.0	+1.21%
60	49	29 400	6.25 mm	1.09	38572.9	+0.38%
80	71	56 800	4.29 mm	0.74	38452.1	+0.06%
100	97	97 000	3.12 mm	0.54	38427.6	—

The buckling load converges monotonically, and it does so as a function of $\Delta x \beta$: the error falls below one per cent once the axial element is about as long as the boundary layer, $\Delta x \simeq 1/\beta$, and below one tenth of a per cent at $\Delta x \simeq 0.75/\beta$. The run times on one core were 24, 65, 91, 150 and 323 seconds.

The post-buckling coefficient behaves completely differently. On the same five meshes b_{1111} was

$$-2.6 \times 10^{-3}, \quad +1.2 \times 10^{-2}, \quad +1.9 \times 10^{-3}, \quad +3.7 \times 10^{-4}, \quad +5.0 \times 10^{-5}$$

changing sign and two orders of magnitude. This is not a mesh resolution problem and no refinement repairs it; the cause is the buckling spectrum of this particular design, treated next.

12.4 The same design with a non-linear pre-buckling state

Repeating the study with `NLprebuck=True` changes both numbers and behaviour:

n_y	n_x	P_{cr} [N]	ΔP_{cr}	n	b_{1111}	Δb_{1111}
30	23	35087.8	+24.59%	4	-0.0565	-62.7%
45	37	28517.4	+1.26%	7	-0.1791	+18.4%
60	49	28309.7	+0.52%	7	-0.1564	+3.3%
80	71	28161.9	—	7	-0.1514	—

Three things change for the better. The buckling load is 27 per cent below the linear-pre-buckling value of Section 12.3, as the edge restraint would lead one to expect. The critical wave number settles at $n = 7$ from $n_y = 45$ upwards instead of wandering. And b_{1111} *converges*: -0.179, -0.156, -0.151, monotonically and without changing sign, to within 3.3 per cent at $n_y = 60$, where the linear analysis of Section 12.3 produced two orders of magnitude of sign-changing noise on the same meshes. The post-buckling coefficient of a CTS cylinder is therefore a computable quantity, but only about a non-linear pre-buckling state.

The coarsest mesh is useless here in a way it was not for the linear analysis: $n_y = 30$ is 25 per cent off in P_{cr} and selects a different mode. That is the boundary-layer requirement of Section 12.2 asserting itself, the expansion point now living where the edge restraint acts.

The default iteration budget is not enough. With the default `NLprebuck_maxiter=12` the $n_y = 45$ run exhausted the budget and stopped at $\lambda_b/\lambda_c = 0.9933$, short of the 0.995 requested, after one Newton-Raphson back-off and two overshoot-and-step-back cycles; the table above was produced with `NLprebuck_maxiter=30`, which converged every mesh with no warning at all. The default was tuned on the constant-stiffness shells of Section 11, where λ_c moves less with λ_b and four to seven load steps suffice. A CTS design with a dense spectrum spends steps backing off, and the budget has to be raised accordingly. The symptom is explicit — the run prints `WARNING: the iterative eigenvalue algorithm stopped at lambda_b/lambda_c = ...` — and must not be ignored, because a state stopped short of the requested neighbourhood is exactly the regime Section 5.3 exists to avoid.

12.5 The buckling spectrum is design-dependent

The circumferential harmonic content of the lowest modes differs qualitatively across the design space. Measured on the $n_y = 60$ mesh with a *linear* pre-buckling state:

design	critical mode	μ_1/μ_0	gap to the next distinct mode
$\theta_{c1} = \theta_{c2} = 45^\circ$	axisymmetric, $n = 0$	1.00018	0.63%
$\theta_{c1} = 30^\circ, \theta_{c2} = 60^\circ, n=3$	axisymmetric, $n = 0$	1.00344	0.34%
$\theta_{c1} = \theta_{c2} = 60^\circ$	pair, $n = 3$	1.00000	14.19%

The $\pm 60^\circ$ design is the textbook situation the single-mode expansion assumes: a clean degenerate pair, well separated from everything above it, and with $n = 3$ the $2n = 6$ harmonic is sampled by ten nodes per wave already at $n_y = 60$.

The other two are not. Their critical mode is *axisymmetric*, and the first six multipliers lie within one per cent of each other, a dense cluster rather than an isolated pair. The single-mode b_{ijkl} is then not determinate, the expansion being made about one member of a cluster the mesh reorders from one refinement to the next, and that is exactly the oscillation of Section 12.3. Worse, the premise of the axisymmetric reduction is violated: the critical mode lies *inside* the subspace the pre-buckling problem is solved in.

The non-linear pre-buckling state removes the pathology. Repeating the steered design with `NLprebuck=True` moves the critical mode off the axisymmetric harmonic entirely: it comes out at $n = 4$ on the $n_y = 30$ mesh and $n = 7$ on both $n_y = 45$ and $n_y = 60$, and b_{1111} becomes -0.179 and -0.156 on the latter two, values of the same order as the reference shells of Section 11 rather than the near-zero noise of the linear case. The axisymmetric critical mode is therefore an artefact of expanding about the *linear* pre-buckling state, in which the edge restraint has not yet developed, and not a property of the design. This is the same separation of the spectrum documented in Section 6.3 for the constant-stiffness shells, seen here in its sharpest form.

The practical consequence is a warning about screening. A spectrum computed with `NLprebuck=False` does not predict the one computed with `NLprebuck=True`, neither the wave number nor the clustering, so the check recommended in Section 12.6 has to be run with the same setting as the production analysis. And if a design genuinely does retain an axisymmetric critical mode under `NLprebuck=True`, the failure is silent: the load stepping halts just short of the bifurcation point, where the reduced tangent is still regular, and reports convergence with no indication that the state it stopped at is about to bifurcate *within* the subspace it was computed in. The diagnostic is the harmonic content of the critical mode, never the convergence history.

12.6 Recommendation

1. **Always inspect the spectrum before using b_{ijkl} .** Run once with `koiter_num_modes=0`, which returns immediately after the eigenvalue analysis and costs a small fraction of a full run, and check (i) that the critical mode is a non-axisymmetric degenerate pair and (ii) that the next distinct multiplier is separated by appreciably more than `NLprebuck_eps1`. If either fails, the buckling load is still meaningful but the single-mode b_{ijkl} is not, whatever the mesh. Run this screening with the *same* `NLprebuck` setting as the production analysis: Section 12.5 shows the two give different wave numbers and different clustering on the same design.
2. **For the buckling load,** size the axial mesh by the boundary layer: $\Delta x \leq 1/\beta$ for about half a per cent, $\Delta x \simeq 0.75/\beta$ for about a tenth. For the test-suite geometry that is $n_y = 60$ and $n_y = 80$ respectively, with `nxt` chosen to hold the aspect ratio near one. Use the finer of the two with `NLprebuck=True`, where the boundary layer carries the expansion point.
3. **For b_{ijkl} , when the spectrum permits it at all,** take $n_y \simeq 8n$ with n the measured circumferential wave number, so that the $2n$ harmonic gets four nodes per wave. For the $\pm 60^\circ$ design, $n = 3$, this is satisfied from $n_y = 24$ upwards, and the mesh is then governed by the buckling load requirement instead.
4. **Do not transfer the 240 elements of the reference.** The 240 S4R elements around the circumference used for the Abaqus cross-check in the reference are a linear reduced-integration shell requirement. The BFSC element is C^1 with cubic interpolation and ten degrees of freedom per node and resolves the same harmonic with far fewer elements; $n_y = 60$ to 80 is the corresponding range here.
5. **Raise `NLprebuck_maxiter`.** The default 12 is a constant-stiffness figure. Use 30 for CTS designs, and treat the `WARNING` about stopping short of $1 - \text{NLprebuck_eps1}$ as a failure rather than a note: the expansion is then made about a state that is not in the neighbourhood of the bifurcation point, which is the very thing Section 5.3 is for.
6. **Cost.** A full run at $n_y = 80$ takes of the order of minutes with `NLprebuck=False` and several times that with `NLprebuck=True`, the load stepping solving an eigenvalue problem per step. In a design-of-experiments loop over thousands of samples, as the reference

performs, this is the binding constraint, and the screening run of item 1 is the part worth doing on every sample.

13 Known limitations

1. **Degenerate critical modes.** Condition 27 supplies one equation per retained mode while the null space is larger, and the rotation inside the degenerate pair still moves b_{1111} by sixteen per cent on the $n_y = 40$ mesh. The canonicalisation of Section 6 makes the choice reproducible rather than arbitrary; it does not make it converged. Multi-mode Koiter is the consistent treatment, and Section 6.3 sizes it: six to eight modes, three to four degenerate pairs, for the reference shells.
2. **Resolution of the second-order field.** The second-order field carries the harmonic $2n$, which the meshes used here sample with about two nodes per wave. This, and not the expansion itself, is what limits b_{ijkl} ; the buckling load is far less affected.
3. **Symmetrisation of ϕ_{30} for distinct modes.** The identity of Eq. 26 was established for coinciding mode indices, which covers the single-mode case exercised here; the general multi-mode symmetrisation remains to be worked out.
4. **Axisymmetric critical modes.** The reduction of the pre-buckling problem to the axisymmetric subspace (Section 4) assumes that no buckling mode lies in that subspace. With a *linear* pre-buckling state parts of the CTS design space violate it, the critical mode there being axisymmetric (Section 12.5). On every design tried, switching to `NLprebuck=True` moved the critical mode off the axisymmetric harmonic and restored the assumption, so the violation has not been observed where the reduction is actually relied upon — but it has not been *excluded* either, and it would fail silently, the load stepping halting just short of the bifurcation point where the reduced tangent is still regular and reporting convergence. No part of the code inspects the harmonic content of the critical mode, which is the only reliable diagnostic.
5. **Load control.** The scheme of Section 5.3 now reaches the requested neighbourhood of the bifurcation point on all meshes tried, but it remains load controlled. The extrapolated eigenvalue problem of Sun Eq. (27), $[\mathbf{K}_T + (\lambda_c - \lambda_b) \partial \mathbf{K}_T / \partial \lambda] \delta \bar{\mathbf{u}} = \mathbf{0}$, would remove the need to sit on top of the bifurcation point at all.
6. **Quadrature.** Convergence with respect to the number of Gauss points has been verified: $n_{int} = 4$ gives b to within 0.4 per cent of $n_{int} = 5$ and 6, while $n_{int} = 2$ under-integrates and produces a rank-deficient element.

A Reproducing the numbers

Every figure in this document that is not a literature value is produced by a script kept in the repository, so that the document and the code cannot drift apart. Two kinds of script are involved. The *verification studies* of Appendix A.1 sweep meshes and report; they take minutes to tens of minutes and assert nothing. The *tests* of Appendix A.2 assert, run under `pytest`, and are the ones that fail if a change breaks something.

All of them require `bfscylinder` $\geq 0.6.0$, whose tangent stiffness matrix is consistent with the internal force vector; several of the figures quoted here measure exactly that consistency, and every one of them changes on earlier versions (Section 11).

A.1 Verification studies

In `doc/verification/`. Each puts the repository root first on `sys.path`, so an installed copy of the package cannot shadow the working tree, and each is run directly, for instance

```
python doc/verification/kinematics_vs_element.py
```

script	produces	runtime
<code>kinematics_vs_element.py</code>	Section 7	seconds
<code>tangent_consistency.py</code>	Section 3	seconds
<code>cts_vs_constant_stiffness.py</code>	Section 12	~10 min
<code>cts_mesh_convergence.py</code>	table of Section 12.3	~12 min
<code>cts_mesh_convergence.py nl</code>	table of Section 12.4	~35 min
<code>cts_mode_spectrum.py</code>	table of Section 12.5	~20 min
<code>reference_mode_cluster.py</code>	table of Section 6.3	~5 min

Two of these deserve a note. `kinematics_vs_element.py` rebuilds f_{int} and the second variation from the strain measures the Koiter expansion assumes and compares them against the compiled element; it includes a deliberately wrong control, the Sanders element driven with von Karman kinematics, which fails at 7.8×10^{-4} and so establishes that the agreement at 10^{-16} is not vacuous. `cts_mesh_convergence.py` runs with `NLprebuck_maxiter=30` rather than the default, for the reason given in Section 12.4.

A.2 Literature verification cases

In `tests/`, run with `pytest` from that directory. The reference values and the meshes they belong to are documented in the tests themselves; the regression values are for their mesh and are not converged, which the comments there state case by case.

test	shell	reference
<code>test_koiter_cylinder_newton_raphson.py</code>	Sun §3.1; AW-CYL-1-1	Sun et al. [6]; Arbocz et al. [1]
<code>test_koiter_cylinder_Waters.py</code>	Waters shell	Arbocz and Starnes (2002)
<code>test_koiter_cylinder_Waters_sanders.py</code>	Waters shell	Arbocz and Starnes (2002)
<code>test_koiter_cylinder_CTS.py</code>	CTS, constant-stiffness limit	dos Santos and Castro [2]
<code>test_koiter_cylinder_CTS_sanders.py</code>	CTS, constant-stiffness limit	dos Santos and Castro [2]
<code>test_buckling_mode_cluster.py</code>	Sun §3.1	Section 6.3
<code>test_linBuck_VAFW.py</code>	VAFW cylinders	linear buckling
<code>test_Zhihua_error.py</code>	VAFW cylinder	linear buckling
<code>test_cts_shares_nlprebuck_algorithm.py</code>	—	source parity

The last of these compares the CTS and `newton_raphson` modules character by character over the regions that implement the algorithm of this document. The algorithm was spliced from one into the other rather than retyped, and that test is what keeps the two from diverging silently; the mesh generation, the per-integration-point ABD of a variable-stiffness laminate, and the CTS-specific commentary are deliberately excluded from the comparison.

References

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